

Chains of reasoning - Set 2 - Solutions

Specification Prf2, Prf3, Prf5.

Answer key

Question	Answer	Topic
1	E	Chains of reasoning
2	F	Chains of reasoning
3	D	Chains of reasoning
4	D	Chains of reasoning
5	H	Chains of reasoning
6	A	Chains of reasoning
7	E	Chains of reasoning
8	B	Chains of reasoning
9	E	Chains of reasoning
10	H	Chains of reasoning
11	F	Chains of reasoning

Worked solutions

Question 1 RSN-09 · Chains of reasoning · spec Prf5

A rectangle has integer side lengths and its area (in cm^2) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting $a \times b$ and $b \times a$ once)?

Answer: E

$$ab = 2a + 2b \Rightarrow (a - 2)(b - 2) = 4: (a - 2, b - 2) = (1, 4) \text{ or } (2, 2), \text{ giving } 3 \times 6 \text{ and } 4 \times 4.$$

Full method - how and why

Turn a two-variable Diophantine equation into a product by the 'add a constant and factorise' trick.

- 1 Set up: $ab = 2a + 2b$ with a, b positive integers.

why: Area equals perimeter, numerically.

- 2 Rearrange: $ab - 2a - 2b = 0$, then add 4 to both sides: $ab - 2a - 2b + 4 = 4$.

why: The added constant is the product of the two subtracted coefficients, chosen to make the left side factorise.

- 3 Factorise: $(a - 2)(b - 2) = 4$.

why: Check by expanding: $ab - 2a - 2b + 4$ ✓.

- 4 Positive factor pairs of 4 with $a - 2 \leq b - 2$: (1, 4) and (2, 2), giving 3×6 and 4×4 .

why: Negative factors would need $a < 2$; $a = 1$ gives $b - 2 = -4$, i.e. $b = -2$, not a length.

- 5 So **two** rectangles.

why: Check: 3×6 has area 18 and perimeter 18 ✓; 4×4 has 16 and 16 ✓.

Where it goes wrong

- Counting 3×6 and 6×3 separately - the question says to count each rectangle once.
- Forgetting to add 4 and being unable to factorise.
- Including the factor pair (4, 1) as a third solution.

Takeaway $ab + pa + qb = r$ factorises after adding pq : $(a + q)(b + p) = r + pq$. Then list factor pairs.

Question 2 RSN-11 · Chains of reasoning · spec Prf5

The product of three consecutive positive integers is N , and the middle one of the three integers is m . Which of the following statements must be true?

- I N is divisible by 6.
- II N is divisible by 4.
- III $N + m$ is a perfect cube.

Answer: F

Among three consecutive integers there is a multiple of 2 and a multiple of 3 (I). II fails for $1 \cdot 2 \cdot 3 = 6$. III: $N + m = (m - 1)m(m + 1) + m = m^3 - m + m = m^3$.

Full method - how and why

Three consecutive integers: automatic divisibility by 2 and 3, but not by 4; plus a neat algebraic identity.

- 1 Write the three as $m - 1, m, m + 1$, so $N = (m - 1)m(m + 1)$.
why: Centring on the middle term is what makes III come out.
- 2 I: among any three consecutive integers there is at least one multiple of 2 and one multiple of 3, so $6 \mid N$. **True.**
why: Every second integer is even and every third is a multiple of 3.
- 3 II: test the smallest case $1 \times 2 \times 3 = 6$, which is not divisible by 4. **False.**
why: Three consecutive integers contain only one or two evens; when only one, it may contribute a single factor of 2.
- 4 III: $N + m = (m^2 - 1)m + m = m^3 - m + m = m^3$ - a perfect cube. **True.**
why: $(m - 1)(m + 1) = m^2 - 1$ is the difference of squares; the $-m$ and $+m$ cancel.
- 5 So I and III.
why: Match to the list.

Where it goes wrong

- Accepting II by testing $2 \times 3 \times 4 = 24$ only - one confirming case is not enough.
- Expanding N the long way instead of using $(m - 1)(m + 1) = m^2 - 1$.

Takeaway k consecutive integers are always divisible by $k!$ - here $3! = 6$, but not necessarily by 4.

Question 3 RSN-12 · Chains of reasoning · spec Prf5

How many positive integers less than 1000 are divisible by neither 2 nor 5?

Answer: D

Of 1 to 999: 499 multiples of 2, 199 of 5, 99 of 10. Divisible by 2 or 5: $499 + 199 - 99 = 599$. Neither: $999 - 599 = 400$.

Full method - how and why

Inclusion-exclusion: count what you don't want and subtract. Take care with 'less than 1000'.

1 The range is 1 to 999, so 999 integers.

why: 'Less than 1000' excludes 1000 itself.

2 Multiples of 2: $\lfloor \frac{999}{2} \rfloor = 499$. Multiples of 5: 199. Multiples of both, i.e. of 10: 99.

why: Divide and round down. 10 is the LCM of 2 and 5.

3 Divisible by 2 or 5: $499 + 199 - 99 = 599$.

why: Subtract the overlap, which was counted twice.

4 Neither: $999 - 599 = 400$.

why: Sanity check: in each block of 10 consecutive integers exactly 4 (ending in 1, 3, 7, 9) survive, and $999 = 99$ full blocks plus 991-999, giving $99 \times 4 + 4 = 400$ ✓.

Where it goes wrong

- Using 1000 as the count and getting 400 by luck with wrong intermediate figures - or getting 399 by an off-by-one at the top.
- Forgetting to subtract the multiples of 10, giving $999 - 698 = 301$.

Takeaway $|A \cup B| = |A| + |B| - |A \cap B|$, with the intersection counted via the LCM; then subtract from the total.

Question 4 RSN-13 · Chains of reasoning · spec Prf5

p and q are prime numbers with $p + q = 45$. Find pq .

Answer: D

An odd sum of two primes needs one of them to be 2 (the only even prime): $q = 43$, which is prime. $pq = 86$.

Full method - how and why

Parity applied to primes: the sum of two primes is odd only if one of them is 2.

1 45 is odd, so one prime is even and one odd.

why: odd + odd = even, even + even = even; only odd + even gives an odd total.

2 The only even prime is 2, so $p = 2$ and $q = 43$.

why: 2 is the unique even prime - every other even number has 2 as a proper factor.

3 Check 43 is prime: not divisible by 2, 3, 5 and $7^2 = 49 > 43$, so yes.

why: Test primes up to the square root only.

4 $pq = 2 \times 43 = 86$.

why: The question asks for the product, not the primes.

Where it goes wrong

- Trying odd pairs like $2 + 43$ versus $5 + 40$ at random; 40 is not prime and no odd pair can work at all.
- Answering 43 or 45 instead of the product.

Takeaway An odd sum of two primes forces one of them to be 2. Parity settles most prime puzzles instantly.

Question 5 RSN-15 · Chains of reasoning · spec Prf5

A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?

- I The sum of all 10 terms is negative.
- II The sequence contains at least one negative term.
- III The sequence contains at least one positive term.

Answer: H

Terms 1-5 and 6-10 are two blocks of five, each with negative sum (I). A five-block has negative sum, so some term is negative (II). A three-block has positive sum, so some term is positive (III).

Full method - how and why

Group the terms into blocks that the given conditions control, then read off each claim.

- 1 I: split the ten terms into terms 1-5 and terms 6-10 - two blocks of five consecutive terms.
why: Each block has negative sum by hypothesis, and the two blocks together are the whole sequence.
- 2 Negative + negative = negative, so the total is negative. **True.**
why: The grouping is exact - no term is used twice or left out.
- 3 II: terms 1-5 sum to a negative number, so at least one of them is negative (five non-negative terms could not). **True.**
why: A sum is negative only if some summand is.
- 4 III: terms 1-3 sum to a positive number, so at least one is positive. **True.**
why: Same argument the other way round.
- 5 So all three.
why: Match to the list. (Such sequences do exist for 10 terms - the conditions become contradictory only for longer sequences.)

Where it goes wrong

- Trying to construct an explicit sequence; the claims follow from the block structure alone.
- Thinking the sum of all ten is ambiguous because 10 is not a multiple of 3 - use the blocks of five, which do fit.

Takeaway Partition the sequence into blocks the hypotheses control; a negative sum forces a negative term, a positive sum a positive one.

Question 6 RSN-17 · Chains of reasoning · spec Prf5

Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?

Answer: A

If Alice is truthful then Ben lies, so Cara is truthful: two truth-tellers, impossible. If Cara is truthful then Alice and Ben both lie; but Alice lying means Ben is truthful: contradiction. So Ben is the truth-teller: Cara lies (indeed Ben is not lying) and Alice lies (Ben is not lying), which is consistent.

Full method - how and why

A truth-teller puzzle. Assume each person in turn is the sole truth-teller and check the whole scenario for consistency.

- 1 Suppose **Alice** is truthful. Then Ben lies, so 'Cara is lying' is false, so Cara is truthful. That makes two truth-tellers - contradiction.
why: Exactly one truth-teller is the constraint; two breaks it.
- 2 Suppose **Cara** is truthful. Then Alice and Ben both lie. But Alice lying means 'Ben is lying' is false, so Ben is truthful - contradiction.
why: Cara's own statement collapses under its consequences.
- 3 Suppose **Ben** is truthful. Then Cara lies, so Ben is not lying ✓ consistent. Alice says 'Ben is lying', which is false, so Alice lies ✓. Cara says both Alice and Ben lie - false, since Ben is truthful ✓.
why: Every statement's truth value matches the assumed roles, with exactly one truth-teller.
- 4 So Ben is the truth-teller.
why: Two cases eliminated, one consistent - the answer is determined.

Where it goes wrong

- Stopping at the first scenario that seems to work without checking every statement.
- Declaring the situation impossible after two failed cases instead of testing the third.
- Forgetting that a liar's statement must come out **false**, which is itself information.

Takeaway Test each candidate exhaustively: a consistent assignment must satisfy every statement **and** the count of truth-tellers.

Question 7 RSN-19 · Chains of reasoning · spec Prf5

A bag contains red and blue balls only. If one red ball is removed, one quarter of the remaining balls are red. If instead two blue balls are removed, one third of the remaining balls are red. How many balls are in the bag?

Answer: E

With r red and b blue: $r - 1 = \frac{1}{4}(r + b - 1)$ and $r = \frac{1}{3}(r + b - 2)$. The second gives $b = 2r + 2$; substituting into the first, $4r - 4 = 3r + 1$, so $r = 5$, $b = 12$: 17 balls.

Full method - how and why

Two conditions, two unknowns: translate each sentence into an equation, then solve simultaneously.

- 1 Let r be red and b blue. Removing one red: $r - 1 = \frac{1}{4}(r + b - 1)$.
why: The total drops by 1 as well as the red count - both sides of the fraction change.
- 2 Removing two blues: $r = \frac{1}{3}(r + b - 2)$.
why: Here the red count is unchanged; only the total falls, by 2.
- 3 From the second: $3r = r + b - 2$, so $b = 2r + 2$.
why: Solve the simpler equation first and substitute.
- 4 Substitute into the first: $4(r - 1) = r + (2r + 2) - 1 = 3r + 1$, so $4r - 4 = 3r + 1$ and $r = 5$.
why: Multiply through by 4 to clear the fraction before substituting.
- 5 Then $b = 12$ and the total is 17.
why: Check: remove one red $\rightarrow$ 4 red of 16 $= \frac{1}{4} \checkmark$; remove two blue $\rightarrow$ 5 red of 15 $= \frac{1}{3} \checkmark$.

Where it goes wrong

- Forgetting that removing a ball also reduces the **total**, giving $r - 1 = \frac{1}{4}(r + b)$.
- Answering r or b rather than the total.
- Not checking the answer against both original statements.

Takeaway When items are removed, adjust the total as well as the category; then verify against both conditions.

Question 8 RSN-20 · Chains of reasoning · spec Prf5

n is a positive integer such that n^2 has exactly 3 positive divisors. Which of the following must be true?

Answer: B

A number with exactly 3 divisors is the square of a prime; so $n^2 = p^2$ and $n = p$ is prime.

Full method - how and why

Use the structure of the divisor count. Exactly three divisors is a very restrictive condition.

- 1 A number with exactly 3 divisors must be p^2 for a prime p : its divisors are $1, p, p^2$.
why: If a number had two distinct prime factors it would have at least 4 divisors; p^3 has 4; p alone has 2.
- 2 So $n^2 = p^2$, and since both are positive, $n = p$.
why: Take the positive square root.
- 3 Therefore n is **prime**.
why: That is the only guaranteed property.
- 4 Check the others: $n = 2$ is even, $n = 3$ is odd - so neither parity is forced; and n prime is never a perfect square.
why: Both $n = 2$ and $n = 3$ satisfy the condition (4 and 9 have three divisors each), which kills the parity options.

Where it goes wrong

- Answering ' n is a perfect square' by confusing n with n^2 .
- Answering $n = 3$ from one example; $n = 2, 5, 7$ all work too.

Takeaway Exactly 3 divisors $\Leftrightarrow$ the square of a prime. Divisor counts come from the prime factorisation exponents.

Question 9 X2-26 · Chains of reasoning · spec Prf5

The positive integers a , b and c satisfy $a < b < c$ and $a + b + c = 12$. Which of the following statements must be true?

- I $a \leq 3$
- II $c \geq 5$
- III $b = 4$

Answer: E

Since $a < b < c$, $3a < 12$ so $a \leq 3$ (I); $3c > 12$ so $c \geq 5$ (II). III fails: $1 + 2 + 9 = 12$ has $b = 2$.

Full method - how and why

Extremal reasoning: squeeze a and c using the ordering, then test the remaining claim with an example.

1 I: since $a < b < c$, we have $3a < a + b + c = 12$, so $a < 4$, i.e. $a \leq 3$ for an integer. **True.**

why: Replacing b and c by the smallest value a gives a strict under-estimate of the sum.

2 II: similarly $3c > a + b + c = 12$, so $c > 4$, i.e. $c \geq 5$. **True.**

why: Replacing a and b by the largest value c over-estimates the sum.

3 III: must $b = 4$? Try $1 + 2 + 9 = 12$ with $1 < 2 < 9$ ✓ and $b = 2$. **False.**

why: One valid triple with $b \neq 4$ settles it.

4 So I and II.

why: Match to the list.

Where it goes wrong

- Writing $3a \leq 12$ instead of $3a < 12$; the strict ordering $a < b < c$ gives a strict inequality, though here both routes end at $a \leq 3$.
- Assuming b must be the mean of the three - true only for equally spaced values.

Takeaway Bound an ordered sum by replacing every term with the smallest (or largest): $3a < a + b + c < 3c$.

Question 10 RSN-02 · Chains of reasoning · spec Prf5

The integers a, b, c satisfy $a + b + c = 0$ and $abc = 12$. Which of the following statements must be true?

- I Exactly one of a, b, c is positive.
- II $a^2 + b^2 + c^2$ is even.
- III One of a, b, c is divisible by 3.

Answer: H

$abc > 0$ means either none or exactly two of them are negative; they cannot all be positive with sum 0, so exactly two are negative and one positive (I). $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = -2(ab + bc + ca)$ is even (II). 3 is prime and divides abc , so it divides one of them (III). (Example: $-1, -3, 4$.)

Full method - how and why

Three claims about integers constrained by a sum and a product. Sign analysis, an algebraic identity, and a prime-divisibility fact.

- 1 I: $abc = 12 > 0$, so the number of negatives is even: 0 or 2. If none were negative, three positives could not sum to 0. So exactly two are negative and one positive. **True.**
why: A positive product means an even count of negative factors; note 0 is impossible since $abc \neq 0$.
- 2 II: use $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$. With $a + b + c = 0$, $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.
why: The identity converts a sum of squares into twice an integer.
- 3 That is $2 \times$ an integer, hence **even. True.**
why: $ab + bc + ca$ is an integer because a, b, c are.
- 4 III: 3 is prime and $3 \mid 12 = abc$, so 3 divides one of the factors. **True.**
why: Euclid's lemma: a prime dividing a product divides one of the factors. This would fail for a composite like 4.
- 5 So all three. (Check with $-1, -3, 4$: sum 0 ✓, product 12 ✓.)
why: A concrete example confirms the constraints are satisfiable at all.

Where it goes wrong

- Assuming exactly one negative because the product is positive - that would make the product negative.
- Trying to settle II by testing examples rather than using the identity.
- Thinking III needs a specific example; it follows from primality alone.

Takeaway $a + b + c = 0$ makes $a^2 + b^2 + c^2 = -2(ab + bc + ca)$; and a prime dividing a product must divide a factor.

Question 11 RSN-18 · Chains of reasoning · spec Prf5

How many ordered pairs of positive integers (m, n) satisfy $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$?

Answer: F

$4n + 4m = mn \Rightarrow (m - 4)(n - 4) = 16$. Positive factor pairs of 16 as ordered pairs:
 $(1, 16), (2, 8), (4, 4), (8, 2), (16, 1)$: five. (Negative factor pairs would make m or n non-positive.)

Full method - how and why

Clear the fractions, then use the add-a-constant factorising trick and count **ordered** pairs.

1 Multiply $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$ by $4mn$: $4n + 4m = mn$.

why: Clearing denominators turns it into a polynomial equation in integers.

2 Rearrange and factorise: $mn - 4m - 4n + 16 = 16$, so $(m - 4)(n - 4) = 16$.

why: Add $(-4)(-4) = 16$ to both sides to complete the rectangle.

3 m, n positive: if $m - 4$ were negative then $m \in \{1, 2, 3\}$ and $n - 4 = \frac{16}{m-4}$ would make n negative. So both factors are **positive**.

why: Check the negative branch explicitly rather than assuming; e.g. $m = 3$ gives $n - 4 = -16$, $n = -12$ ✗.

4 Positive factor pairs of 16: $(1, 16), (2, 8), (4, 4), (8, 2), (16, 1)$ - **five** ordered pairs.

why: 'Ordered' means (m, n) and (n, m) count separately: $(5, 20), (6, 12), (8, 8), (12, 6), (20, 5)$.

Where it goes wrong

- Answering 3 by counting unordered pairs when the question says ordered.
- Forgetting $(4, 4)$, the square factor pair.
- Including negative factor pairs without checking they give positive m, n .

Takeaway $\frac{1}{m} + \frac{1}{n} = \frac{1}{k}$ becomes $(m - k)(n - k) = k^2$; then count factor pairs, respecting ordered versus unordered.